

DHEERAJ VURUKUTI

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WORK EXPERIENCE

Software Engineer, WSU AgWeatherNet

February 2025 – Present | Pullman, WA

- Owned end-to-end architecture and delivery of AWN CropAI (2.6k+ downloads), a production iOS application built with **SwiftUI**, training ML models on large-scale historical weather datasets to forecast Fruit Surface Temperature and improve grower decision-making.
- Increased imaging accuracy by engineering a computer vision pipeline (**Python, OpenCV, YOLO**) that corrects atmospheric distortion in captured thermal imagery, improving reliability of FST estimation.
- Served as Lead Architect for the flagship AgWeatherNet mobile app (1.3k+ downloads) built with **Flutter**, driving system design, development, deployment, and feature evolution across the full product lifecycle.
- Led frontend modernization of a high-traffic AWN platform (29K+ active users), migrating legacy codebases to **React + TypeScript (Vite)** to improve maintainability, bundle efficiency, and long-term scalability.
- Developed AWN Smart Farms, a precision Ag platform (**React + TypeScript, Vite**), enabling geospatial visualization, real-time IoT data processing, automation, and scalable analytics over distributed sensor data.
- Built and launched AWN CropAI-SWD (183+ downloads), a cross-platform **Flutter** application that automates larvae detection using AI-driven image analysis, reducing manual inspection workload for lab and field teams.
- Designed a native **Kotlin** Android app to visualize real-time IoT data streams from distributed devices.

Graduate Research Assistant, Washington State University

April 2023 – December 2024 | Pullman, WA

- Led thesis research on “*Uncertainty-Aware Machine Learning Techniques for Sustainable Malware Detection.*” integrating predictive models with uncertainty quantification to enhance detection accuracy and robustness.
- Developed a **Deep Kernel Learning (DKL) malware detection framework** combining neural networks and Gaussian Processes in PyTorch, achieving a **31% improvement in AUC-F1** over state-of-the-art baselines.
- Mitigated concept drift** by designing a selective-classification strategy leveraging uncertainty quantification to abstain on low-confidence predictions, improving robustness in adversarial and evolving environments.
- Validated model reliability through extensive ablation studies, isolating uncertainty quantification as the primary contributor to performance gains under real-world distribution shifts.

Graduate Teaching Assistant, Washington State University

August 2022 – December 2024 | Pullman, WA

- Mentored students in *Cyber Security and Cryptography*, *Web Development*, and *Professional Skills in Computing and Engineering* courses by conducting interview style evaluations and individualized office hours.

AWARDS & RECOGNITION

- AE50 Award Winner (2026)** – ASABE: Recognized AWN CropAI among the **top 50 global innovations** in product and systems technology.

PROJECTS

Transformer-Based Malware Analysis — *PyTorch, Hugging Face (BERT)*

- Built a BERT-based model on Smali code achieving **0.92 AUC**, maintaining robustness under dataset drift using active learning and conformal prediction.

Object Detector iOS App — *SwiftUI, CoreML, Firebase*

- Developed an iOS app performing on-device ResNet-based object detection with live camera inference and Firebase-backed user authentication.

Human Activity Recognition — *TensorFlow, CNN-LSTM*

- Achieved **89.8% accuracy** by modeling spatiotemporal sensor patterns with a hybrid **CNN-LSTM**

TECHNICAL SKILLS & TOOLS

- Programming & Scripting:** Python, C, C++, Java, R, Swift, Kotlin, Dart, TypeScript, SQL
- AI/ML:** TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face, OpenCV, MediaPipe, YOLO, NLTK
- Web & Mobile:** SwiftUI, Flutter, React, React Native, Node.js, HTML, CSS, REST APIs
- Infrastructure & Tools:** Git, Docker, Linux, CUDA, MongoDB, Roboflow, DBeaver, GA4

EDUCATION

Washington State University | Masters in Computer Science (Thesis – AI/ML)

December 2024 | Pullman, WA

Andhra University | Bachelors in Computer Science

May 2022 | Andhra Pradesh, India